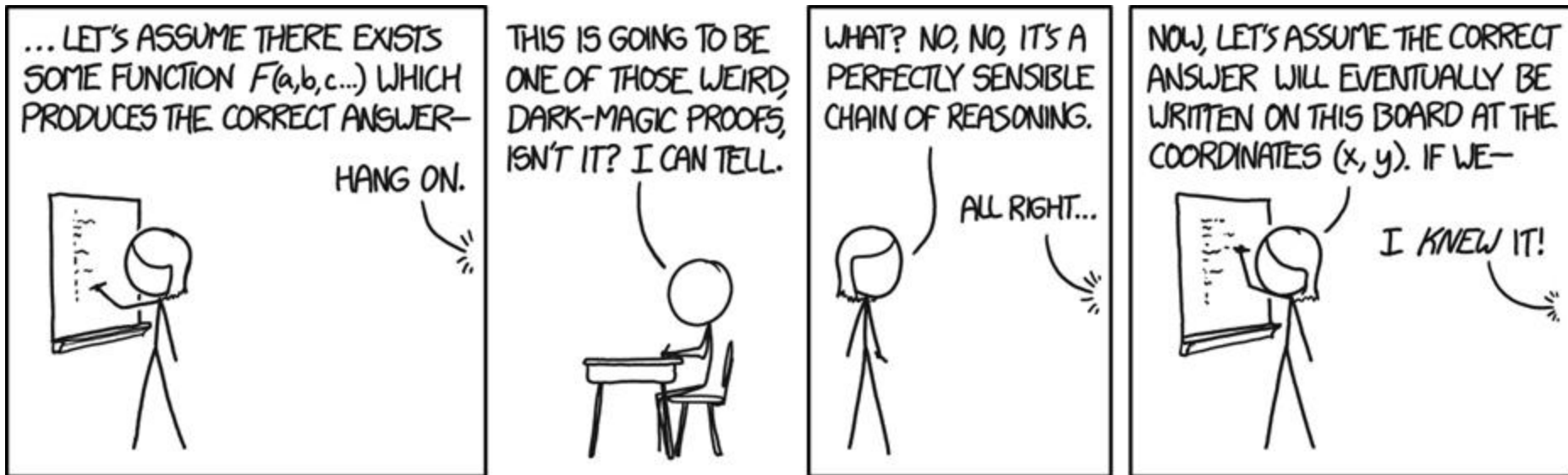


Diagonalization, Undecidability, and the Barber Paradox



Review: Decidability

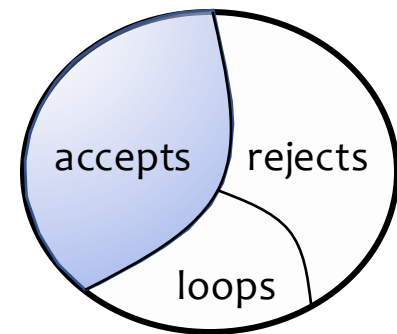
Question: What are the possible outcomes of a TM **M** on an input **x**?

Answer: **M** either (i) accepts, (ii) rejects, or (iii) it “**loops**” (**forever**)

Definition: A Turing Machine **M** **decides** a language **L** if it:

1. accepts every string in **L**, and
2. rejects every string not in **L**

(and does not loop forever on any input)



A language **L** is **decidable** if there is a TM that decides **L**.

*“there is an algorithm that solves the problem associated with **L**”*

Otherwise **L** is **undecidable**.

*“there is no algorithm that solves the problem associated with **L**!”*

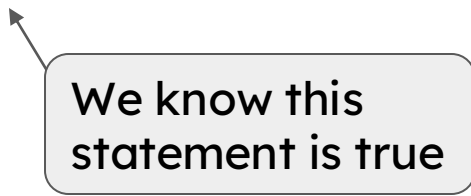
Review: Church–Turing Thesis

“Any natural notion of being ‘algorithmically computable’ is captured by Turing Machines.”

This is not a theorem,
but everyone seems to believe it.
We will assume it’s true and proceed from there.

Anything that can be computed by Python, C++, a quantum computer, LaTeX, pseudocode, AI, etc. can be computed by a Turing Machine.

“Turing-Complete”



We know this
statement is true

Question: To prove that a language L is decidable, must we write down all the details of an actual TM?

No b/c TM is equivalent to pseudocode

Question: If a language L is decidable, then must $L \cup \{\epsilon\}$ also be decidable?

Let M be a machine that decides L

Define $M'(x)$:

If $x = \epsilon$: accept

Else: run $M(x)$ and return the same answer

Existence of Undecidable Languages

To show that there **exists** an **undecidable** language (i.e. a language that no TM decides), we would like to show that:

total # TMs < total # languages.

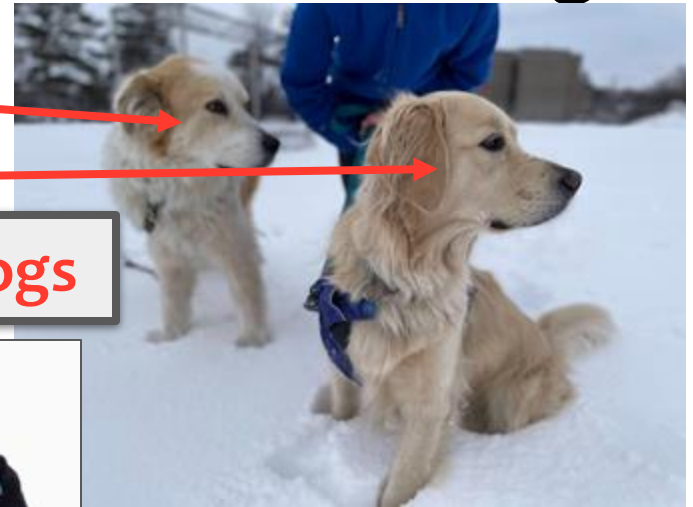
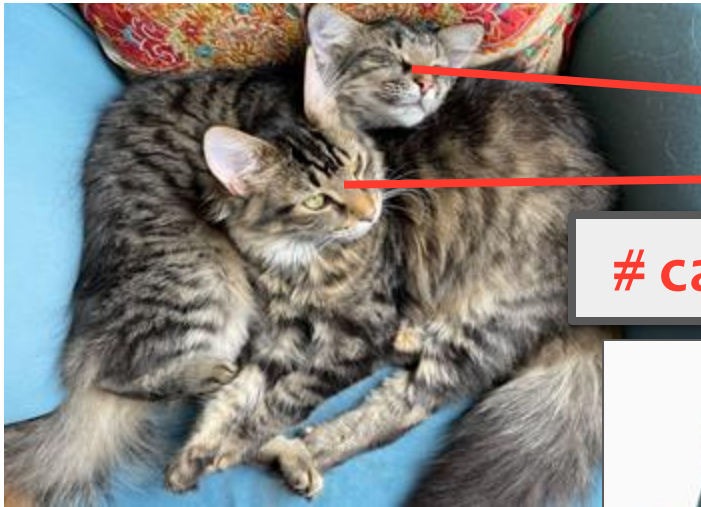
Each TM decides ≤ 1 language

But both of these quantities are **infinite** so what does that even mean?

To infinity and beyond!



Chris Peikert has these Cats and Dogs

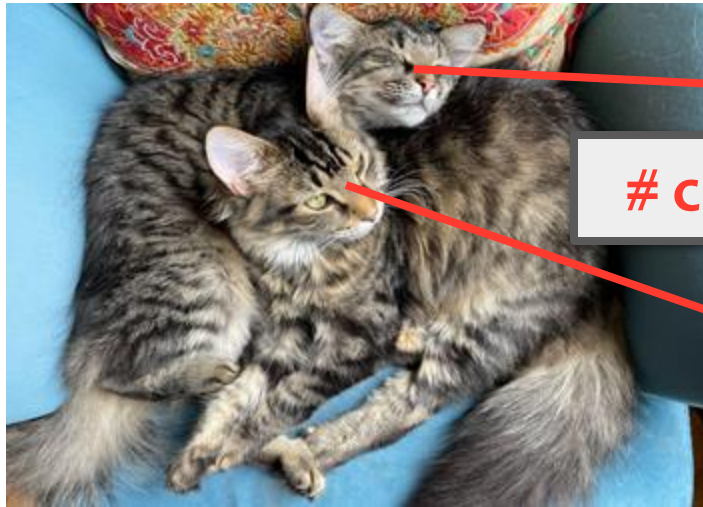


cats $\leq$ # dogs

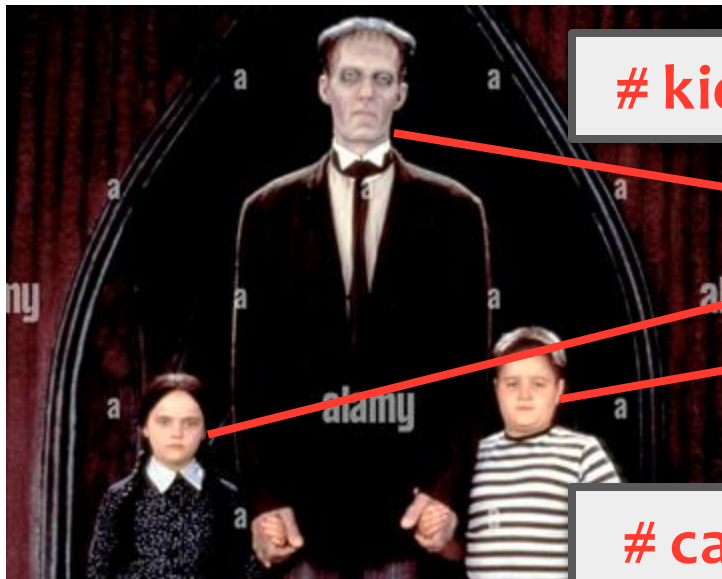
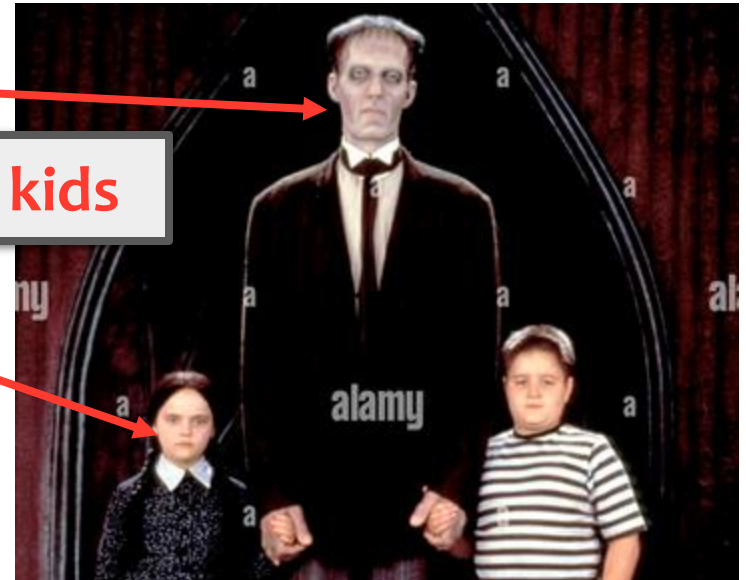


dogs $\leq$ # cats

Chris Peikert has these Cats and Kids



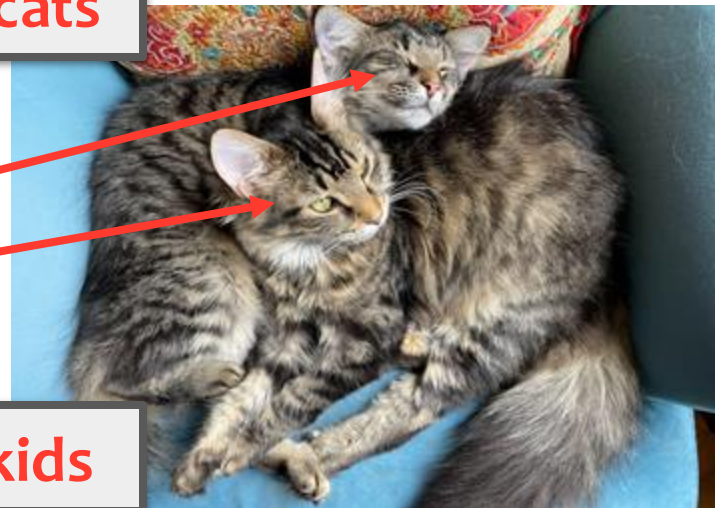
$\# \text{ cats} \leq \# \text{ kids}$



$\# \text{ kids} \not\leq \# \text{ cats}$

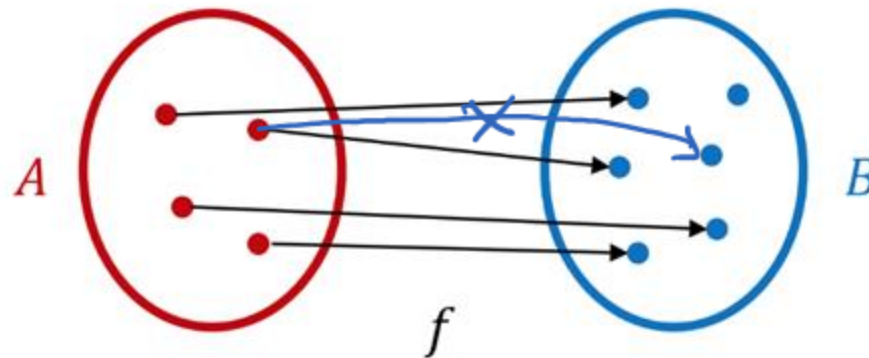
??

$\# \text{ cats} < \# \text{ kids}$



203 Review: Functions and Set Cardinality

- * A (total) function $f: A \rightarrow B$ maps each element $a \in A$ to an element $f(a) \in B$.
- * A function is **1-to-1 (injective)** if each element $a \in A$ is mapped to a different element $f(a) \in B$.
 - * Formally: $\forall a_1, a_2 \in A. a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$



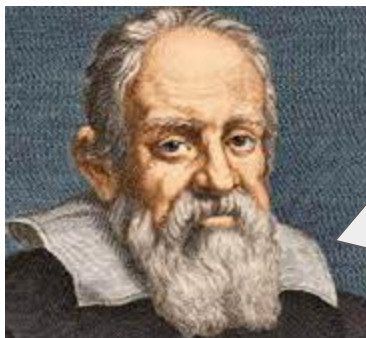
“cardinality of A”



- * If a 1-to-1 function $f: A \rightarrow B$ exists, then we say $|A| \leq |B|$.

Warning! Properties of “ $\leq$ ” for *finite* values might not apply for *infinite* ones!

203 Review: Countability



Galileo (1638)

The attributes
“equal,” “greater,”
and “less,” are not
applicable to
infinite, but only to
finite, quantities.



Actually
yes they
are

Georg Cantor (1895)

- * **Definition:** A set S is **countable** if there's a 1-to-1 mapping from S to the naturals $\mathbb{N} = \{0, 1, 2, \dots\}$, i.e. $|S| \leq |\mathbb{N}|$.
- * **Equivalently:** S is countable if and only if there is a list in which every element appears: $S = \{s_1, s_2, s_3, \dots\}$.

Is ____ a countable set?

The integers from 1 to 10?

Every finite set is countable

The non-negative even numbers?

0, 2, 4, 6, 8, 10, ...

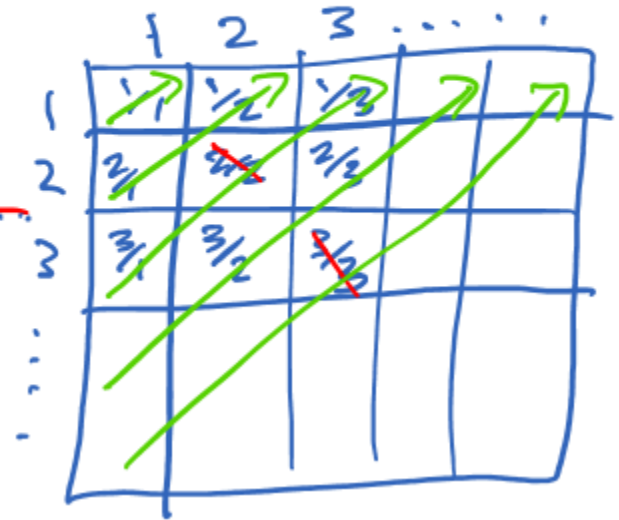
The integers?

~~*Attempt 1: 0, 1, 2, 3, ..., 1, 2, 3, ...*~~

The positive rationals?

0, 1, -1, 2, -2, 3, -3, ...

The reals in $(0,1)$?



Real Number in $(0,1)$

Natural Number

.1	1
.2	2
.3	3
...	...
.11	11
.12	12
...	...

Check out my
proof that the
reals in $(0,1)$
are countable!



There are at least
two different flaws
here



- irrational #'s not on list!
↳ infinite length but naturals have finite length
- leading 0s e.g. 0.0001 not on list

Diagonalization



Theorem: the set of reals in $(0,1)$ is *uncountable*.

Suppose for contradiction we could write a list of all the reals in $(0,1)$.

$r_1 = .59407123\dots$
 $r_2 = .01102876\dots$
 $r_3 = .31454280\dots$
 $r_4 = .10999875\dots$
 $r_5 = .50000000\dots$
 $\vdots$

Integers have
finite length

Our goal is to find a
real in $(0,1)$ that's not
in this supposed list!



Real not on list: $.46324\dots$

$0.09999\dots = 0.10000$

Don't choose 0s or 9s

(Why doesn't the same argument work to show that the integers are uncountable?)



Kronecker

I don't know what predominates in Cantor's theory – philosophy or theology, but I am sure that there is no mathematics there.

I believe, and hope, that a future generation will laugh at this hocus pocus.



Wittgenstein

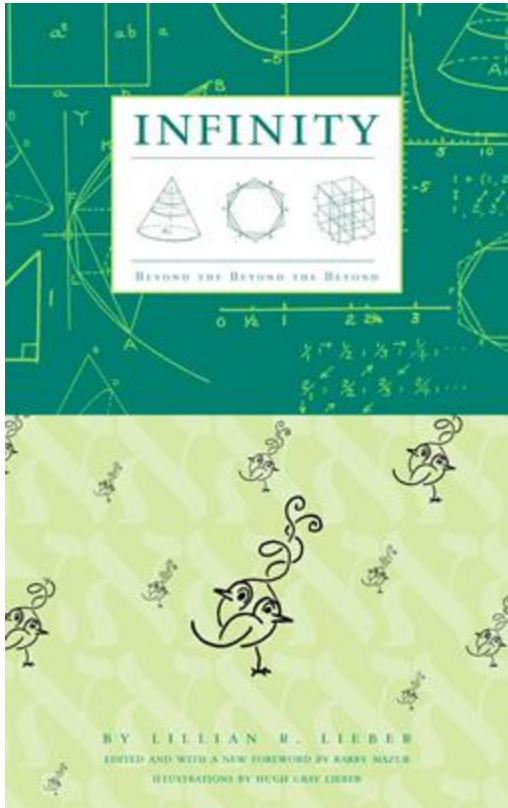
A Bag of Reals

I'm going to reach into
this bag of reals and pick
one out!

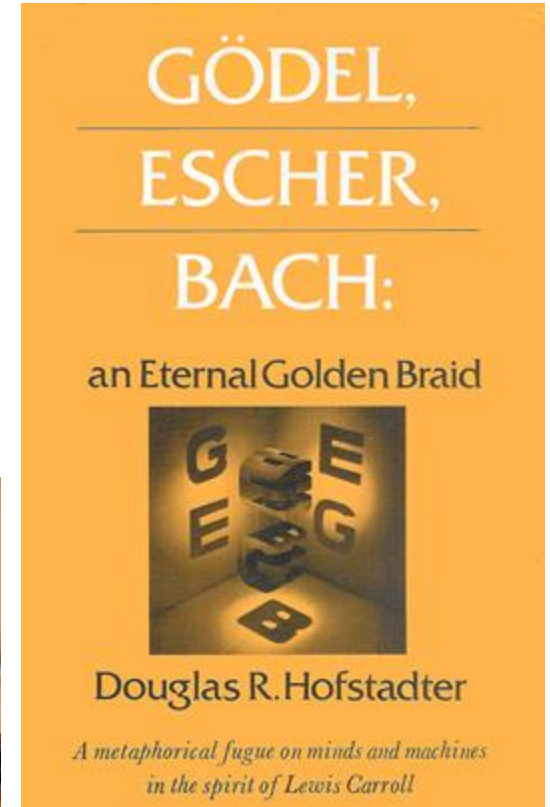


Book Recommendations about Infinities & More

Mark Brehob:



Chris Peikert:



Is ____ a countable set?

The set of all possible TM inputs from $\{0,1\}^*$?
(i.e. the set of all finite-length binary strings)

Countable: $\epsilon, \underbrace{0, 1}_{\text{length 1}}, \underbrace{00, 01, 10, 11}_{\text{length 2}}, \underbrace{\dots}_{\text{length 3}}, \dots$

The set of all TMs?

Countable since each TM can be represented as a finite length binary string.

The set of all languages?

Recall: To show that there **exists** an **undecidable** language we want to show:
total # TMs < total # languages.

Formal Definition of a Turing Machine

A Turing machine is a seven-tuple

$$M = (\Sigma, \Gamma, Q, q_{\text{start}}, q_{\text{acc}}, q_{\text{rej}}, \delta)$$

- Σ is the (finite) *input alphabet*.
- Γ is the (finite) *tape alphabet*, $\Sigma \subseteq \Gamma$
- Q is the *finite set of states*.
- $q_{\text{start}} \in Q$ is the *initial state*.
- $q_{\text{acc}}, q_{\text{rej}} \in Q$ are the (distinct) *accept state* and *reject state*
 $F = \{q_{\text{acc}}, q_{\text{rej}}\}$
- δ is the *transition function*.

All of these
sets are
finite



$$\delta: \underbrace{(Q \setminus F)}_{\text{Non-terminal state}} \times \underbrace{\Gamma}_{\text{Cell contents}} \rightarrow \underbrace{Q}_{\text{Next state}} \times \underbrace{\Gamma}_{\text{Symbol to write}} \times \underbrace{\{L, R\}}_{\text{Where to move the head}}$$



There are uncountably many languages!

Suppose for contradiction there were countably many languages.

We could list them all:

Strings $\longrightarrow$

Languages $\downarrow$

	s1	s2	s3	s4	s5	s6	...
L1	✓	✗	✗	✓	✓	✓	...
L2	✗	✗	✓	✗	✓	✗	...
L3	✓	✓	✓	✓	✗	✓	...
L4	✗	✗	✗	✗	✗	✗	...
L5	✓	✗	✓	✗	✗	✗	...
...	...	...	...	...	...	...	...



A language not in this list: ✗ ✓ ✗ ✓ ✓ ...

A Bag of Languages

Conclusions:

- There are uncountably many languages
- There are countably many *decidable* languages
(since there are countable many TMs, and each TM decides ≤ 1 language)
- There are more languages than decidable languages, so there must be an undecidable language (in fact the VAST majority must be undecidable)

I'm going to reach into
this bag of languages and
pick one out!



What we've shown

We showed that there *exists* an *undecidable* language.

Conclusion: there is a problem that *no computer can solve!*

Next: we'll identify a specific undecidable language!

The Barber Paradox

(aka Russell's Paradox)



“Barbara is the best barber in town! She cuts the hair of those people—and only those people—who do not cut their own hair.”

Question: *does Barbara cut her own hair?*

Barber Paradox is actually Diagonalization in Disguise

Make a table of everyone in town and whose hair they cut:

Hairdressers $\longrightarrow$

Hairdressers $\downarrow$

	P1	P2	P3	P4	P5	...
P1	0	1	1	0	1	...
P2	1	1	0	0	0	...
P3	0	0	0	0	0	...
P4	0	1	1	1	0	...
...	...	...	...	...	...	...
B	1	0	1	0	...	...

Barbara behaves as the flipped diagonal!

Thus, Barbara is not on the list — Barbara does not exist!

Recall: A TM can be represented as a finite string.

WILD IDEA: we can give any TM *description* as
input to any TM!

Using the Barber Paradox to Construct an Undecidable Problem

“**Barbara** *cuts* the hair of all **those**—and only **those**—who do not *cut* their own hair.”

Let’s consider a computational analogy, where:

- **people** = **TM(s)**
- hair = description of TM
- *cut* = *accept*

“**TM** **M_{BARBER}** *accepts* the description of any **TM**—and only such a **TM**—that does not *accept* its own description.”

Does **TM** **M_{BARBER}** accept its own description?

Again it's Diagonalization in Disguise

“**TM** M_{BARBER} *accepts* as input the description of all **TMs**—and only those **TMs**— that do not *accept* as input their own description.”

Make a table of which **TMs** accept as input the description of which **TMs**

	M1	M2	M3	M4	M5	...
M1	0	1	1	0	1	...
M2	1	1	0	0	0	...
M3	0	0	0	0	0	...
M4	0	1	1	1	0	...
...	...	...	...	...	...	...

M _{BARBER}	1	0	1	0	...
---------------------	---	---	---	---	-----

M_{BARBER} behaves as the flipped diagonal, so isn't on the list, so doesn't exist!

⇒ The language $L_{\text{BARBER}} = \{\langle M \rangle : M \text{ does not accept } \langle M \rangle\}$ is undecidable.